1. Purpose

This session was held as part of *Work Package 1 – Requirements Definition* of the **PENTION** project.
The goal was to collect insights, operational challenges, and practical needs from the Dutch Police to refine the ongoing requirements analysis for PENTION.
The meeting followed the "Requirements Elicitation Workshop – LEAs" structure and presentation previously prepared for this purpose.

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2. Opening and Context

We clarified that the meeting's purpose was to **brainstorm and exchange views**, not to conduct a technical review.
The discussion focused on identifying what law enforcement officers **expect and need from the system**, in order to ensure the final solution aligns with real operational contexts.

PENTION was briefly introduced again as a **mobile-lab concept ("Ghostbusters van")**, combining software (AI-based signal analysis) and hardware (sensors, detection units).
The focus was on both **software, hardware, and procedural requirements**, especially concerning inter-agency collaboration.
We structured the discussion around nine key topics:

1. Operational challenges
2. Portability
3. Detection accuracy
4. AI assistance
5. Explainability
6. Substances & precursors
7. Operational context
8. Evidence & data handling
9. User interaction & feedback

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3. Previous Questionnaires

Leonie asked whether other LEAs were also involved in this session.
We clarified that **this workshop was dedicated to the Dutch Police**, while other LEAs had already been queried via questionnaires.

Jos confirmed that he had already **completed two questionnaires** (on 6 and 13 October 2025) covering:

- Operational challenges
- Substances and precursors

These inputs were sent to Damian and Daouiz earlier in October.
However, this information hadn't been circulated to us before the workshop.

As a result, we agreed to **skip topics already covered** in the surveys (operational challenges and substances/precursors) and focus instead on **the remaining aspects**.

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4. Portability and System Configurations

Marco (RIEC) raised an important point: in earlier stages, the requirements mentioned a *handheld device*, yet the discussion had since shifted to a *van-based system*.
He questioned whether the handheld concept was dropped too soon, as it might still have practical value.

Leonie clarified that the current understanding was a **combination** of:

- a **portable sampler** (hand-operated for air intake), and
- a **stationary analysis unit** (installed in or near the van).

Jos added that two configurations are now being discussed:

- **PENTION-S (Stationary)**
- **PENTION-M (Mobile)**
  and that MION is already assessing **power consumption and technical specifications** for the mobile setup.

We all noted that while mobility is a key asset, **cost and feasibility** must also be considered.
It was therefore agreed that **economic feasibility** should be treated as an **additional requirement category** (i.e. "business case" or "cost implication").

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5. Use Cases and Focus

Jos reminded us that the Dutch Police had already defined three operational use cases in the questionnaires:

1. Air measurements in urban areas.
2. Air measurements in postal/distribution hubs.
3. MDMA bulk detection at checkpoints.

We agreed that **use cases should drive the requirements**, not the other way around.

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6. Detection Accuracy

Jos explained that the Dutch Police had already provided feedback on detection requirements:

- **High sensitivity**,
- **Low detection limit**, and
- **High selectivity** to distinguish synthetic drugs and precursors **without false positives**.

Leonie raised an important point: **too much sensitivity can become counterproductive**, detecting irrelevant trace amounts (e.g. residual cocaine on banknotes or reused containers).

She emphasized the need to define **practical detection thresholds** — distinguishing significant signals from background noise or "old positives."

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7. AI Assistance and Explainability

We discussed the two main interpretations of AI support:

1. **Analytical/operational AI** – e.g., pattern detection, trend analysis, or air dispersion modeling to locate emission sources.
2. **Legal/forensic AI** – models whose outputs might be used as part of evidence.

Jos stressed that **AI models must be validated** if their output is to be admissible in court.
Leonie noted that in a previous project (*RITHMS*), the **Public Prosecutor's Office** reacted negatively when involved too late.
She recommended **early engagement** of prosecutors or legal experts to avoid similar issues and ensure the tools are legally acceptable.

We discussed briefly the need for **data contextualization** (e.g., linking geolocated readings to company or environmental data), and Willem-Jan raised potential **GDPR implications**.
It was agreed that for law enforcement, the system should currently be seen as a **supportive tool** providing *indications*, not yet *legal proof*.

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8. User Interaction and Feedback

Jos described how current detection devices (e.g., TruNarc, First Defender, MX908) provide:

- Binary results (*hit / no hit*),
- Substance identification,
- Confidence level (%).

This simple and clear interface is preferred by field officers who are not chemical experts.
We agreed that PENTION should follow a similar user-feedback model: **clear binary results with confidence values** and optional detailed information.

We also discussed **external certification** of devices (e.g., equivalent to *KEMA Keurmerk*) as part of future validation.

Leonie asked about PENTION's added value compared to existing instruments.
Jos clarified that the PENTION system is designed for **large-scale air sampling** — e.g., full containers, palettes, or rooms — something current devices cannot do.

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9. Closing Remarks

We concluded that the meeting was productive despite partial topic overlap with previous surveys.
Several **additional requirements and clarifications** were gathered, especially regarding portability, cost feasibility, AI validation, and detection thresholds.